ODE: $\frac{d^2\phi}{dx^2} + \frac{d^2\phi}{dy^2} = 0$ Since $\phi$ is independent of $y$.

$\therefore$ Given Laplace's Partial differential becomes the ordinary differential equation $\frac{d^2\phi}{dx^2} = 0$.

$\therefore$ Integrating, $\phi(x) = ax + b$.

$\phi(-1) = k_0$, $\phi(1) = k_1$, $a + b = k_1$.

$\Rightarrow -a + b = k_0$, $2b = k_0 + k_1$.

$\therefore a = \frac{-k_0 + k_1}{2}$ $b = \frac{k_0 + k_1}{2}$.

$\therefore \phi(x) = \frac{k_1 - k_0}{2} x + \frac{k_0 + k_1}{2}$.

In the same Problem, determine $\Omega(z)$,

Since $\phi$ satisfies harmonic, we can find the harmonic conjugate $\psi(x)$.

C-R equations, $\frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y} = \frac{k_1 - k_0}{2}$ - (1)

$\frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x} = 0$. - (2)

The second equation indicates that $\psi$ is a function of $y$ alone.

Integrating w.r.t '$y$'

$$\psi(y) = \frac{k_1 - k_0}{2} y.$$

for convenience, taken the constant of integration to be zero.

$\therefore$ complex potential function

$$\Omega(z) = \phi(x) + i\psi(y),$$

$$= \frac{k_1 - k_0}{2} x + \frac{k_1 + k_0}{2} + i \frac{k_1 - k_0}{2} y.$$

<u>Problems</u>: 1) If $w = \phi + i\psi$ represents the complex potential for an electric field and $\psi = x^2 - y^2 + \frac{x}{x^2+y^2}$, determine the function $\phi$.

<u>Solution</u>: $\therefore W = i\left(z^2 + \frac{1}{z}\right) + A.$

$\therefore \phi = -2xy + \frac{y}{x^2+y^2}, \quad \psi = (x^2 - y^2) + \frac{x}{x^2+y^2}.$

2) Given $\phi$ is the velocity Potential for the Planar flow of an incompressible fluid, find the velocity field f of the flow. Also evaluate complex velocity potential of the flow.

$$\phi(x,y) = \frac{x}{x^2+y^2}.$$

Solution:

$$\frac{\partial \phi}{\partial x} = \frac{y^2-x^2}{(x^2+y^2)^2}. \qquad \frac{\partial \phi}{\partial y} = -\frac{2xy}{(x^2+y^2)^2}.$$

$$\therefore f' = \frac{y^2-x^2}{(x^2+y^2)^2} - i\,\frac{2xy}{(x^2+y^2)^2}.$$

$$\therefore \Omega(z) = \frac{1}{z} + C.$$

3) Find the orthogonal trajectories of the family of curves $x^4+y^4-6x^2y^2 =$ constant.

Solution: $u = x^4+y^4-6x^2y^2.$

$$\frac{\partial u}{\partial x} = 4x^3 - 12xy^2. \qquad \frac{\partial u}{\partial y} = 4y^3 - 12x^2y.$$

$$v_y = 4x^3 - 12xy^2 \qquad v_x = 12x^2y - 4y^3$$

$$\therefore v = 4x^3y - 4xy^3 + C \qquad v = 4x^3y - 4xy^3 + C.$$

Complex Potential

$f(z) = \phi(x,y) + i\psi(x,y)$

$\phi$ - velocity potential / Potential fn/.

$\psi$ - Stream function / lines of force / flux function.

1) An electrostatic field in the xy-plane is given by the potential function $\phi = 3x^2y - y^3$, find the Stream function? $\psi = ?$

2) If the potential function $\phi = \log(x^2+y^2)$, find the flux function and complex potential.

$\psi$, $f(z)$ / $\Omega(z)$.

3) In a two-dimensional fluid flow, the Stream function $\psi$ is given, find the velocity potential $\phi$.

i) $\psi = -y/x^2+y^2$.

ii) $\psi = \tan^{-1}(y/x)$.